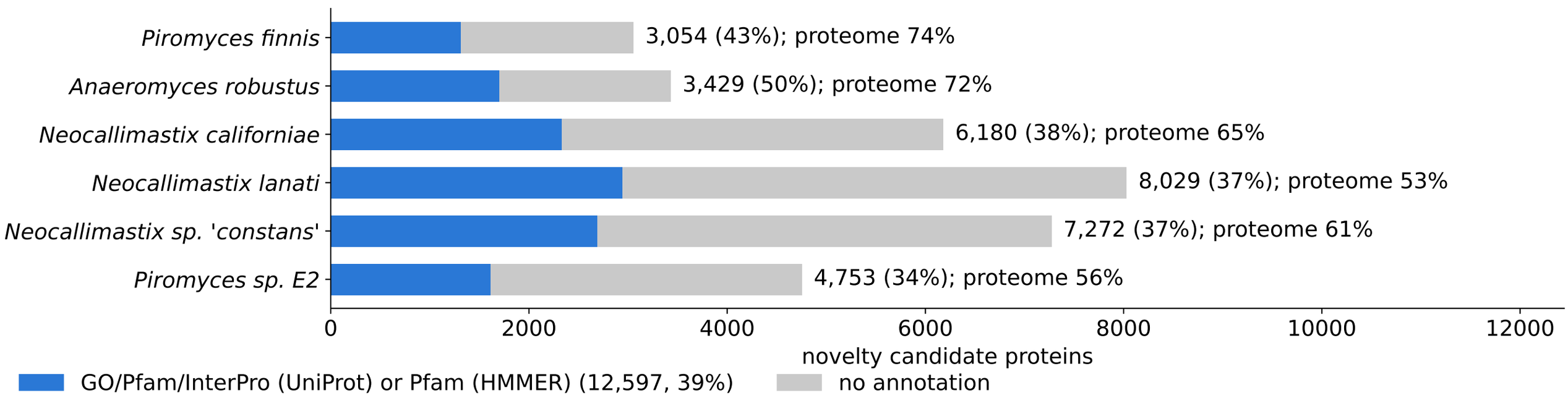
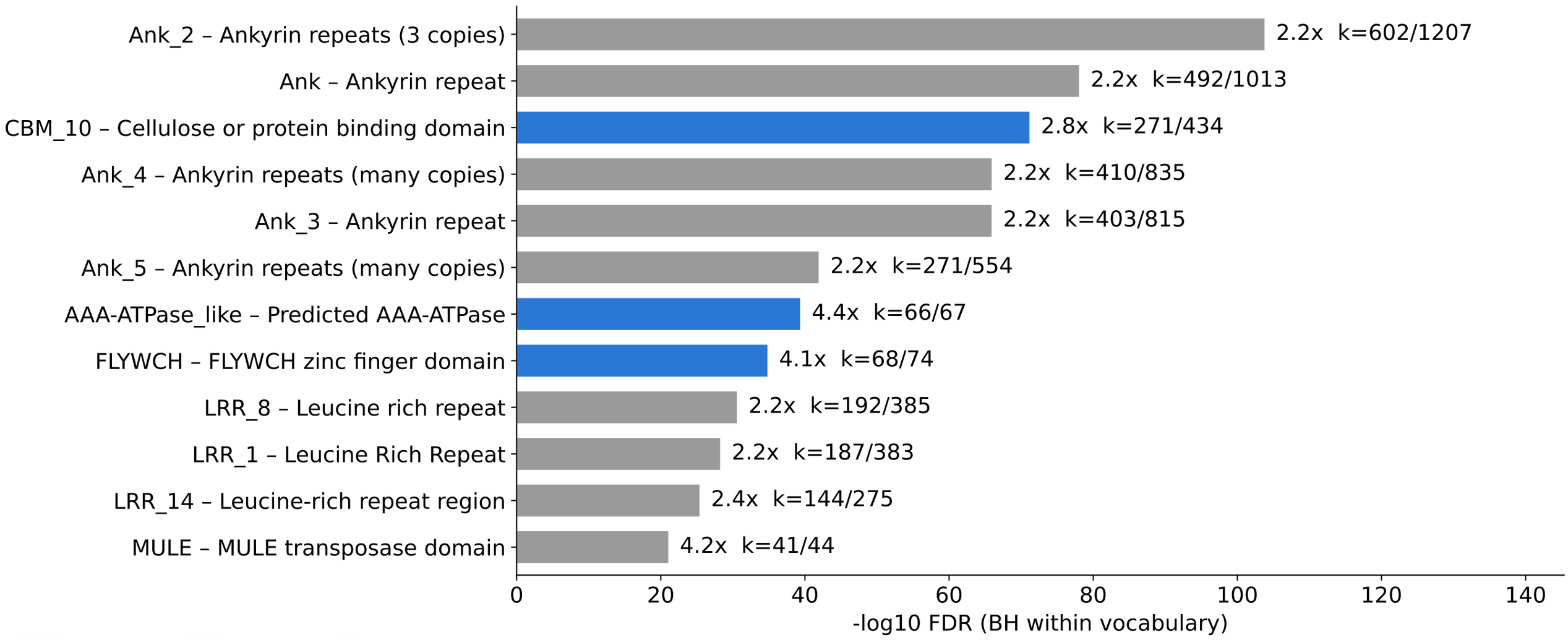
Neocallimastigomycota (6 IN) vs early fungi (13 OUT): enrichment of novelty candidates (present in >=0.75 of IN, absent from all OUT; diamond pairwise, nf_NovInvenio main d03f5a2 run 2026-10-11)

A 32,717 candidates in 6 IN species



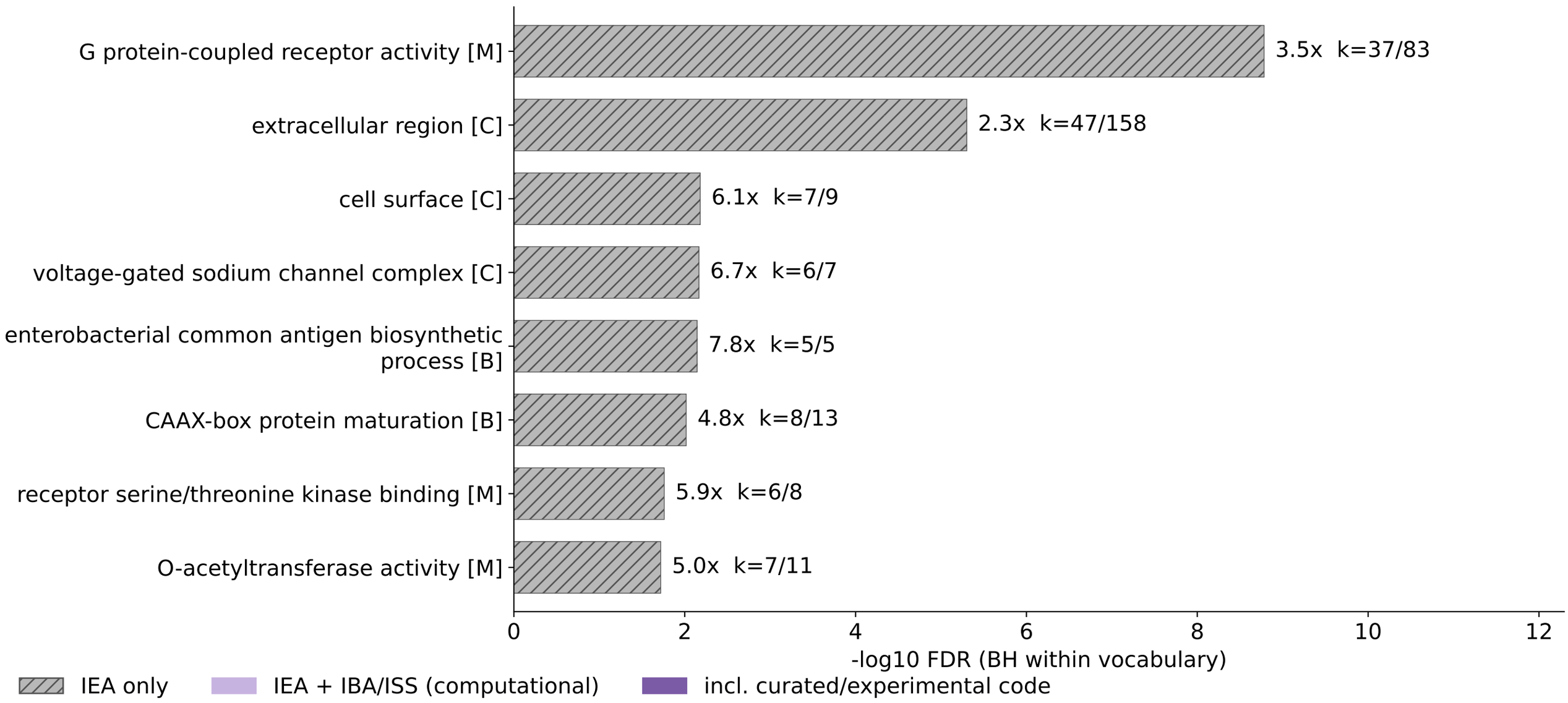
B Pfam families enriched in candidates (primary; family units on both sides)

1045 terms tested; 56 at FDR<0.05; annotated units: 3576 cand. / 16075 all



C GO terms (secondary; UniProt computational evidence)

2553 terms tested; 18 at FDR<0.05; annotated units: 891 cand. / 6975 all



Annotation: UniProt reference proteomes (Pfam, InterPro, GO) for Piromyces finnis and Anaeromyces robustus. Pfam only (Pfam-A 38.2, HMMER 3.4 hmmsearch or hmmscan, --cut_ga; no GO, no InterPro) for the 4 NCBI/JGI proteomes: Neocallimastix californiae, Neocallimastix lanati, Neocallimastix sp. 'constans', Piromyces sp. E2. Pfam and Pfam-clan tests cover 6 of 6 IN species; GO and InterPro tests cover 2 of 6. Universe: all proteins of the 6 IN proteomes. Family units: candidate families from the pipeline; background clustered with mmseqs (0.3 id, 0.8 cov, cov-mode 0). One-sided hypergeometric; BH within each vocabulary. Bars: FDR<0.05 and k>=3 candidate units. Novelty candidates: 32717.